slides tinyurl.com/icap2026-commitment

# *Toward an AI-assisted Comprehensive Meta-analysis of the Organizational Commitment Literature*

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UNIVERSITY OF GUELPH

the SYNTHESIS COMPANY of CALIFORNIA

# Coauthors

The team behind this AI-assisted meta-analysis


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# Agenda

Where we've been, where we are, and where this is going

- Brief History of Commitment Theory & Research
- Challenges to conducting comprehensive reviews
- Toward an AI-Assisted Meta-Analysis
    - Where we are
    - Where we are going
    - What is possible

# Organizational commitment: the early years

A brief history — some milestones

| | |
|---|---|
| 1960s–70s | Multidisciplinary interest in commitment; multiple competing conceptualizations |
| 1979 | Mowday, Steers & Porter introduce the OCQ — the seminal book follows in 1982 |
| 1990 | Mathieu & Zajac publish the first comprehensive meta-analysis |
| 1990–91 | Allen & Meyer introduce the TCM measures (1990) and the three-component model (1991) |
| 2002 | Meyer, Stanley, Herscovitch & Topolnytsky — comprehensive meta-analysis of TCM research |

# Two comprehensive meta-analyses

Studies **(k)** and participants **(N)** per analysis — Mathieu & Zajac (1990) and Meyer et al. (2002)

## 1990

Mathieu & Zajac (1990)

STUDIES PER ANALYSIS (k)

**3–43**

PARTICIPANTS PER ANALYSIS (N)

**302–15,531**

VARIABLES META-ANALYZED 48

26 | 14 | 8

## 2002

Meyer et al. (2002)

STUDIES PER ANALYSIS (k)

**3–69**

PARTICIPANTS PER ANALYSIS (N)

**441–23,656**

VARIABLES META-ANALYZED 38

19 | 10 | 9

antecedents correlates consequences

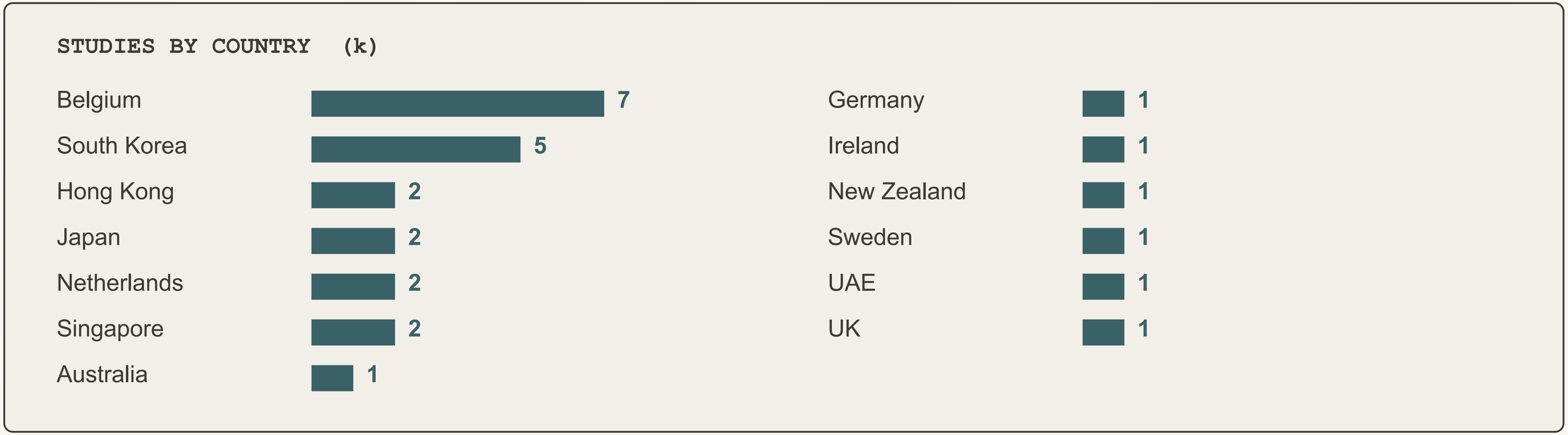
Mathieu & Zajac (1990), Psychological Bulletin, 108(2) · Meyer et al. (2002), Journal of Vocational Behavior, 61

# A modest cross-national base

Meyer et al. (2002) — studies reporting country of origin

GEOGRAPHIC REACH OUTSIDE NORTH AMERICA

**13** countries · **27** studies




Studies reporting country of origin (n = 27); tabulation from the ICAP 2014 presentation.

# Challenges to conducting comprehensive reviews

Why a truly comprehensive commitment review has become impractical by hand.

**1**

### Studies keep growing

Commitment research piles up faster every decade.

**2**

### Reviews turn narrow

Meta-analyses cover fewer variables to stay feasible.

**3**

### Even narrow is too much

Now even targeted reviews outstrip human capacity.

**4**

### AI shares the labour

AI screens and codes at scale, clearing the bottleneck.

# An AI-assisted pipeline — with humans in the loop

Screening and coding were performed by Eva, a family of AI agents; the team set the rules and decided every uncertain case.

| 1 | | 2 | | 3 | | 4 | | 5 |
|---|---|---|---|---|---|---|---|---|
| **Search** | → | **De-duplicate** | → | **Screen** | → | **Code** | → | **Validate** |
| 7 databases, keyword + citation | | consolidate overlapping hits | | every record; flags its own uncertainty | | we created 82 structured fields; AI coded each abstract | | automated consistency checks |

**HUMANS STAY IN THE LOOP**

- Set & refine the inclusion rules
- Decide every case the AI flags as uncertain
- Pilot-review the coding before each full run

**The AI does the volume; the team sets the rules and owns the judgment calls.**

FINDING THE STUDIES

# *Literature Search*

A Global Search

# Literature search method

Two complementary search strategies in every database (except ERIC)

- **Keyword & subject-term search —** truncated organizational terms (org*, work*, employ*, job*, firm*, institution*, affect*, norm*, continu*) within three words of commit*, plus each database's subject headings; syntax adapted to each platform.
- **Cited-reference search —** every record citing any of 14 foundational commitment works (e.g., Porter et al., 1974; Mowday et al., 1979; Allen & Meyer, 1990; Meyer & Allen, 1991; Meyer et al., 2002; Klein et al., 2014).
- **Scholarly sources only —** academic journals, books, dissertations & theses, conference papers, reports, and working papers.
- **All languages —** the search applied no language restriction, so abstracts in any language were captured, not only English.

## Seven databases

Searched December 20, 2025 – January 7, 2026

| | |
|---|---|
| ABI/INFORM Collection | ProQuest |
| Business Source Complete | EBSCO |
| ERIC | Ovid · keyword only |
| Dissertations & Theses | ProQuest |
| APA PsycINFO | Ovid |
| Scopus | Elsevier |
| Web of Science | Clarivate |

# Records identified, by database and search strategy

Records retrieved per database, before consolidation and de-duplication · bars share one scale

| Database | Keyword search | Cited-reference search |
|---|---|---|
| ABI/INFORM Collection | 26,659 | 9,642 |
| Business Source Complete | 29,551 | 6,382 |
| ERIC | 5,372 | — not supported |
| Dissertations & Theses | 17,936 | 20,410 |
| APA PsycINFO | 16,257 | 5,516 |
| Scopus | 25,611 | 19,505 |
| Web of Science | 23,898 | 15,660 |
| **Total** | **145,284** | **77,115** |

Combined: **222,399 records**

ERIC (Ovid) does not support cited-reference searching.

# Retrieved to Coded

De-duplication only removes overlap — AI relevance screening did the real narrowing

**222,399** **records retrieved**
7 databases · keyword + citation

▼ de-duplication — *removes duplicate hits only*

**113,643** **unique records**
after de-duplication

▼ **AI relevance screening by the Eva family of agents** — on each record's title & abstract

**22,484** **coded abstracts**
22,605 relevant;
121 had no abstract to code

**Only ~20% of unique records passed AI relevance screening** (22,605 of 113,643) — the rest were off-topic, not duplicates.

# How accurate is AI screening? — from prior research

### Relative Screening Accuracy

**97.2%**
AI

**91.6%**
expert humans

of relevant studies kept in rather than missed
5 meta-analyses · 30,150 abstracts screened

### Time saving with AI

**≈ 72×**

faster than screening by hand

36 hours of hand screening, done by the AI in under 30 minutes.
One of the five · 2 reviewers + 1 adjudicator · AI: a single pass

### How the accuracy was measured

The AI and the human team each screened every title and abstract, deciding relevant or not. Where the two disagreed, the record was re-checked to establish which studies genuinely belonged. The figure shown is the percentage of those relevant studies a method correctly kept in rather than missed, averaged across the five datasets.

Mehr, A., Howard, J. L., Nouroozi, C., Khorramnazari, B., Banks, G. C., Tay, L., Cuijpers, P., Miguel, C., Harrer, M., Meyer, J. P., Stanley, D. J., Wang, X., Luo, G., Huo, B., Liu, J., & Rousseau, D. M. (2026). *Improving Evidence Synthesis with Artificial Intelligence* [Preprint].

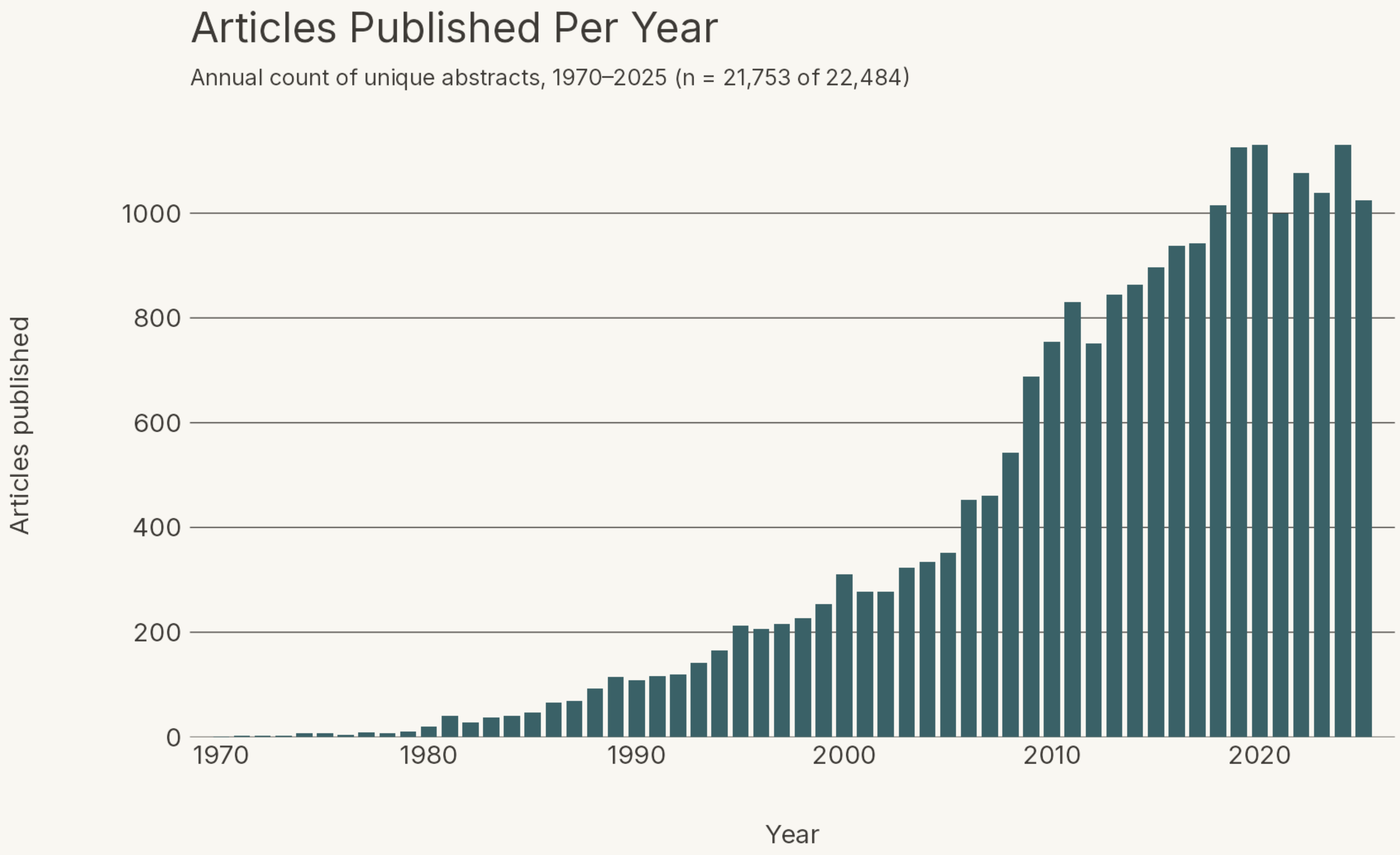
WHAT THE CODED ABSTRACTS REVEAL

# *Abstract Analysis*

How the literature has grown — and what it studies.

How quickly has the literature on organizational commitment grown?

Here's the count of articles published **per year**.

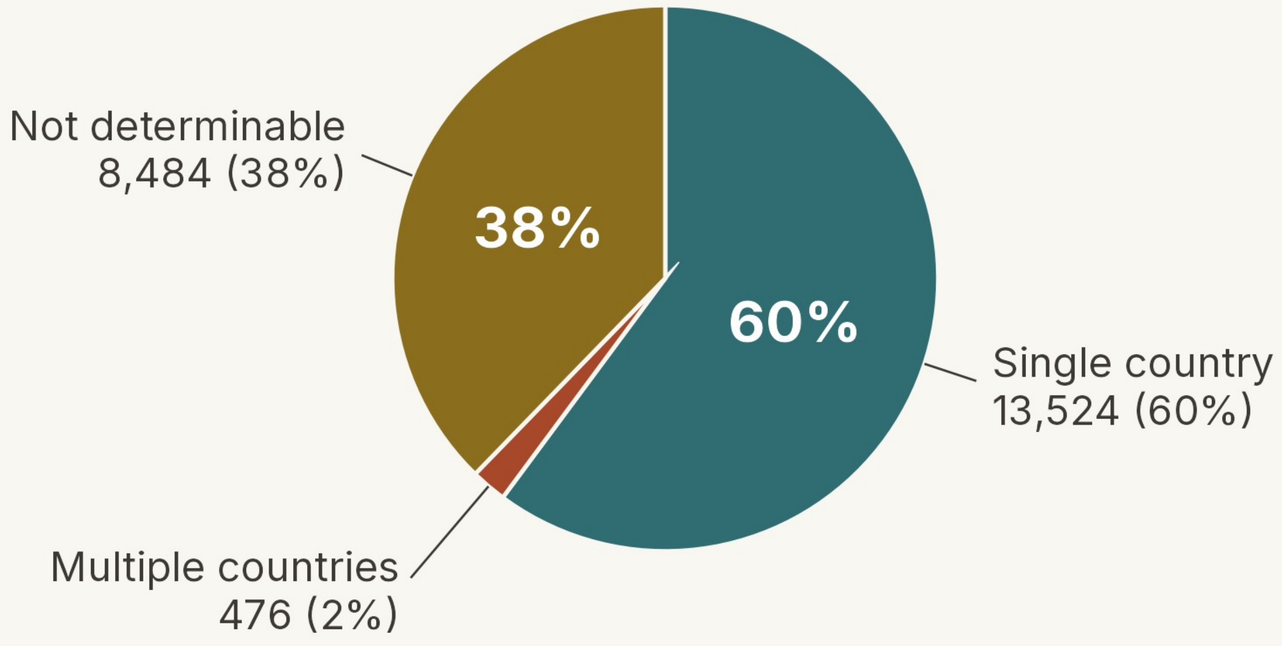

Articles Published Per Year
Annual count of unique abstracts, 1970–2025 (n = 21,753 of 22,484)
Articles published
0
200
400
600
800
1000
1970
1980
1990
2000
2010
2020
Year

[static]

How has research on organizational commitment spread around the world over five decades?

# In what countries were the data obtained?

Country-coding status across all 22,484 unique abstracts



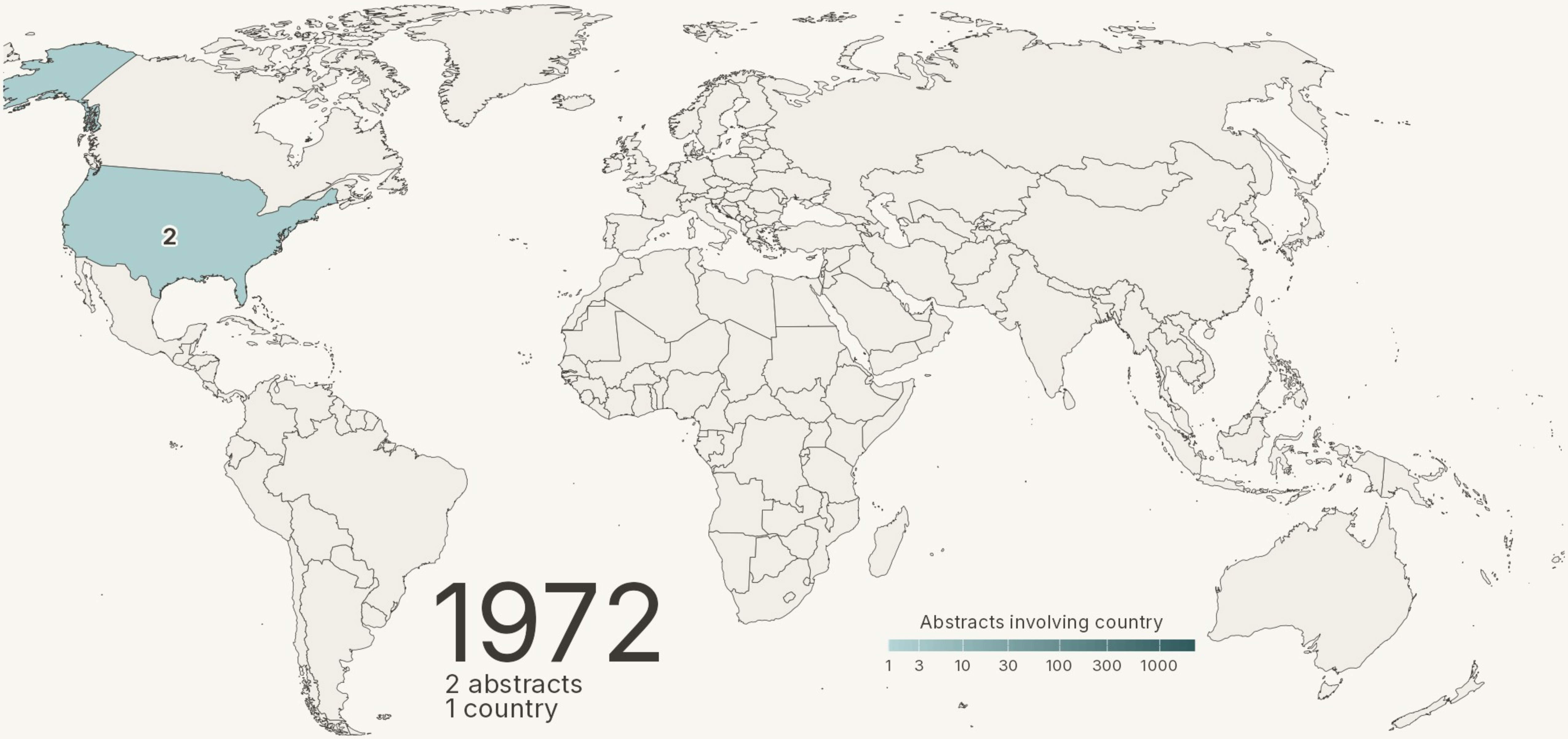

### How country of origin was determined

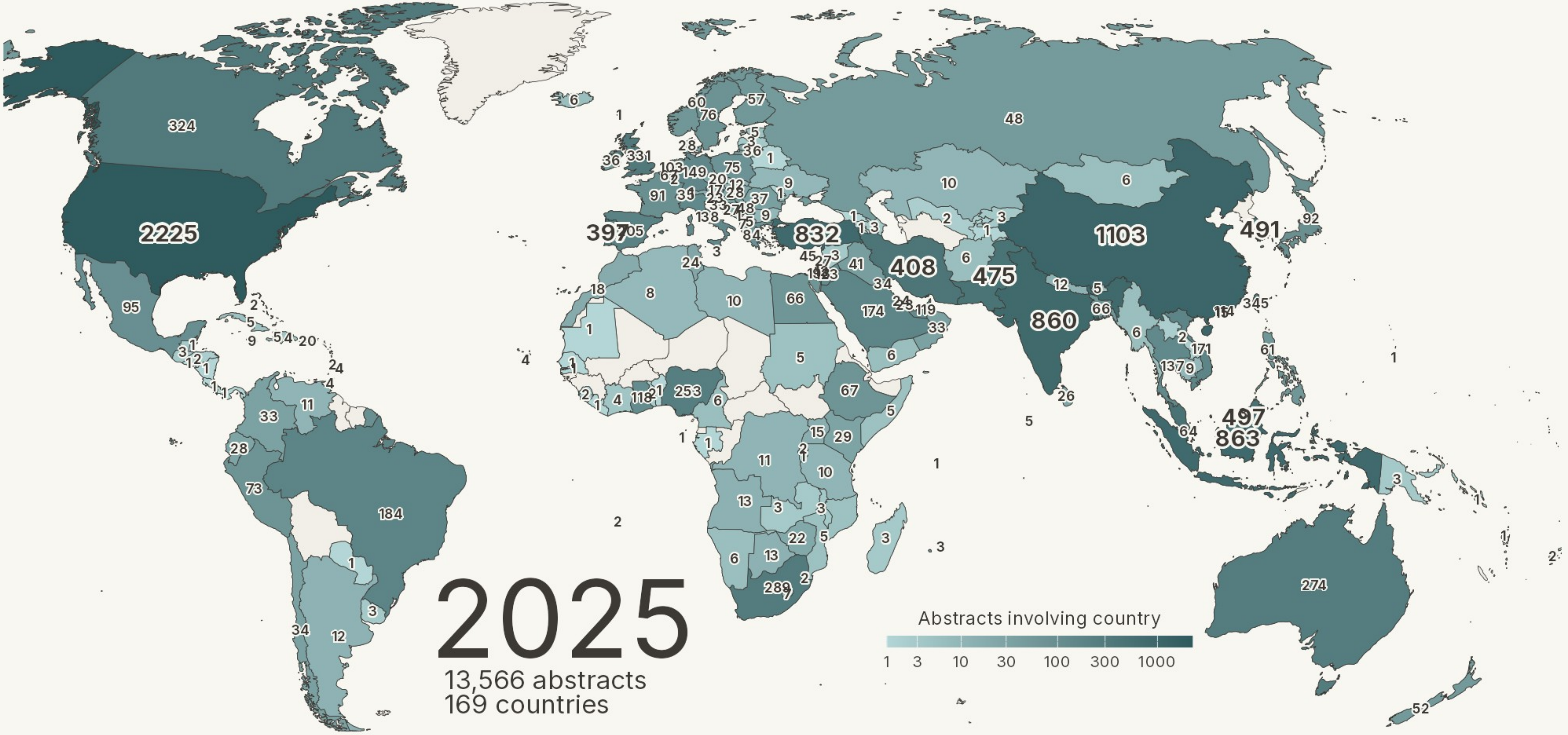
**Country coding criteria (each YES / NO):**

- Is a country named directly in the abstract?
- Does a stated city or region map to a country?
- Does a named company or organization map to a country?
- Do all authors' affiliations point to the same country?

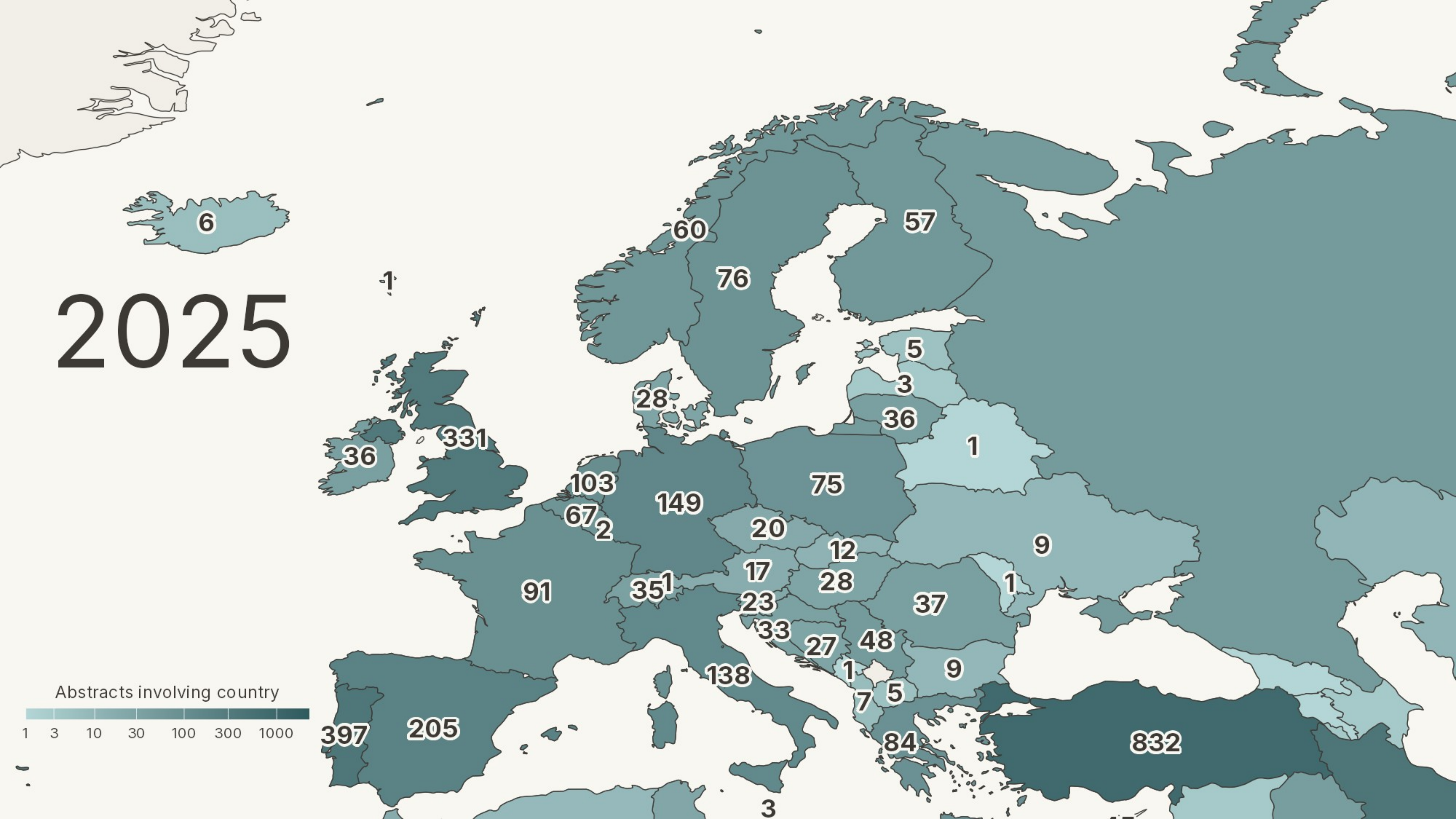
**Recovering undetermined cases:**

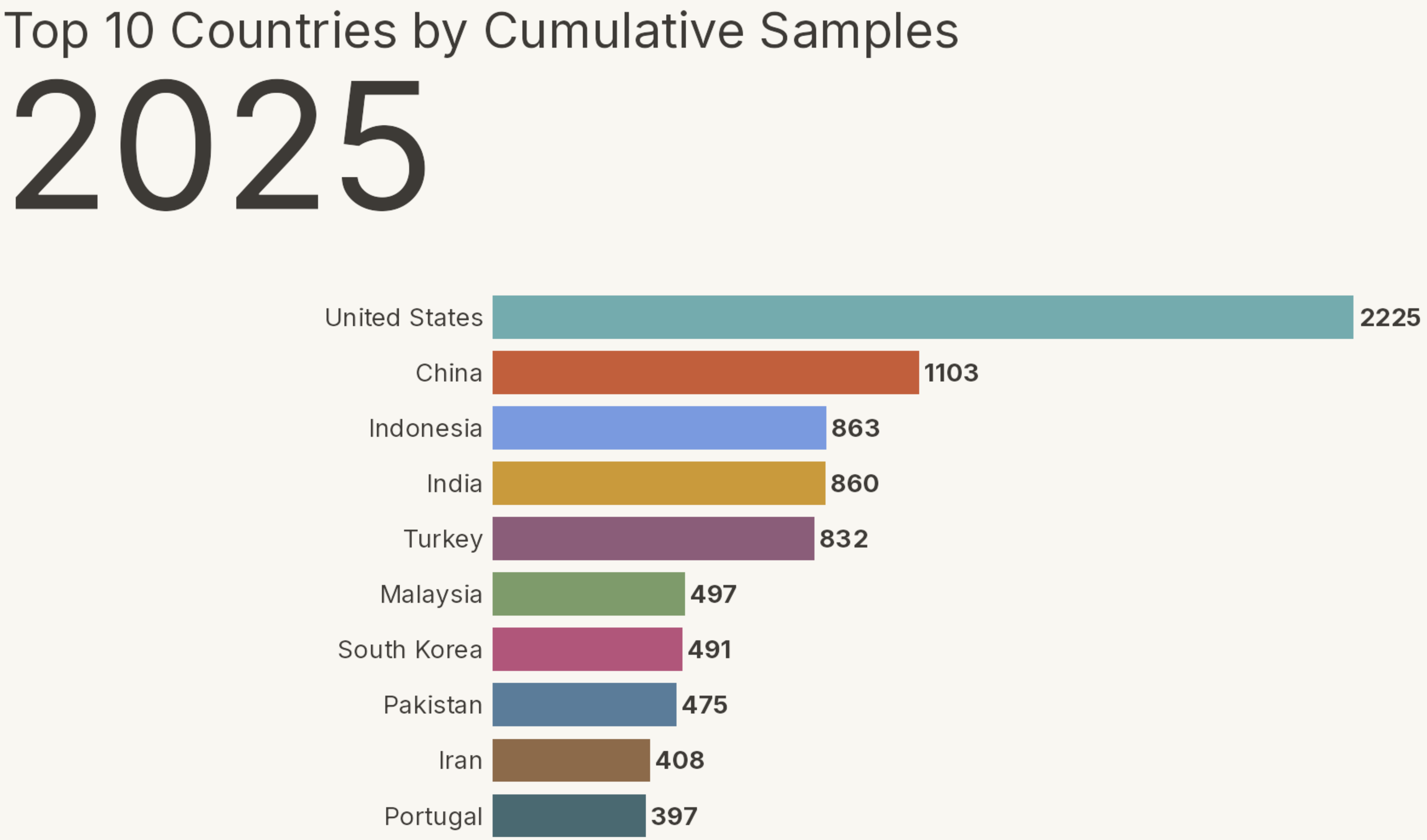
- For undetermined cases, author affiliations were retrieved from OpenAlex via DOI lookup and passed to the model, resolving 14 previously undetermined items.


2
1972
2 abstracts
1 country
Abstracts involving country
1 3 10 30 100 300 1000


2025
13,566 abstracts
169 countries
Abstracts involving country
1 3 10 30 100 300 1000
2225
1103
860
863
832
497
491
475
408
397
324
274
253
184

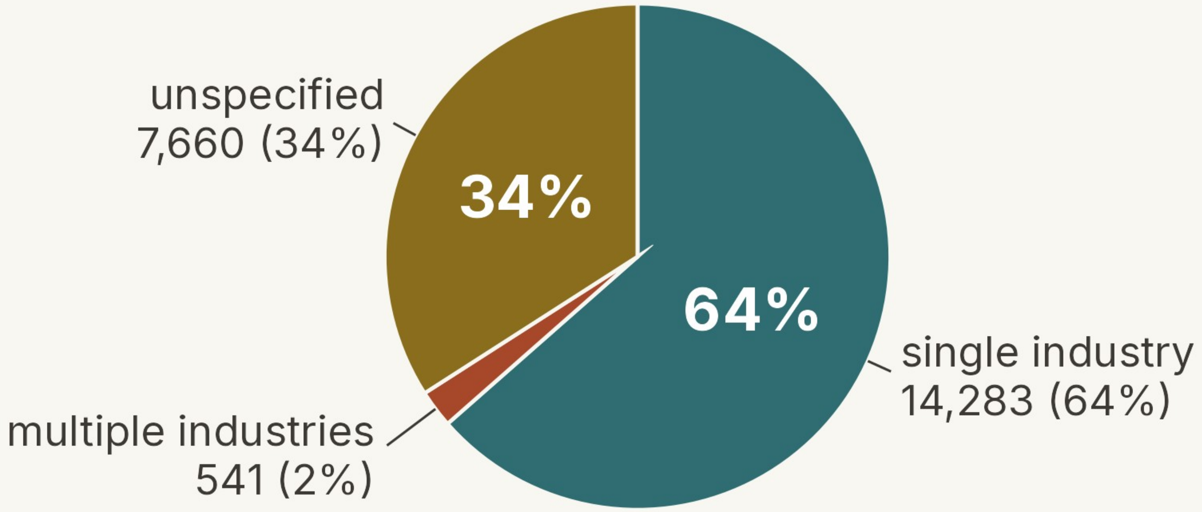
[static]


2025
Abstracts involving country
1 3 10 30 100 300 1000
6
60
57
76
1
5
3
28
36
331
36
1
103
75
149
67
2
20
9
12
17
91
35
1
28
1
23
37
33
27
48
138
1
9
5
7
397
205
84
832
3


Top 10 Countries by Cumulative Samples
2025
United States
2225
China
1103
Indonesia
863
India
860
Turkey
832
Malaysia
497
South Korea
491
Pakistan
475
Iran
408
Portugal
397

# What Industries have been Studied

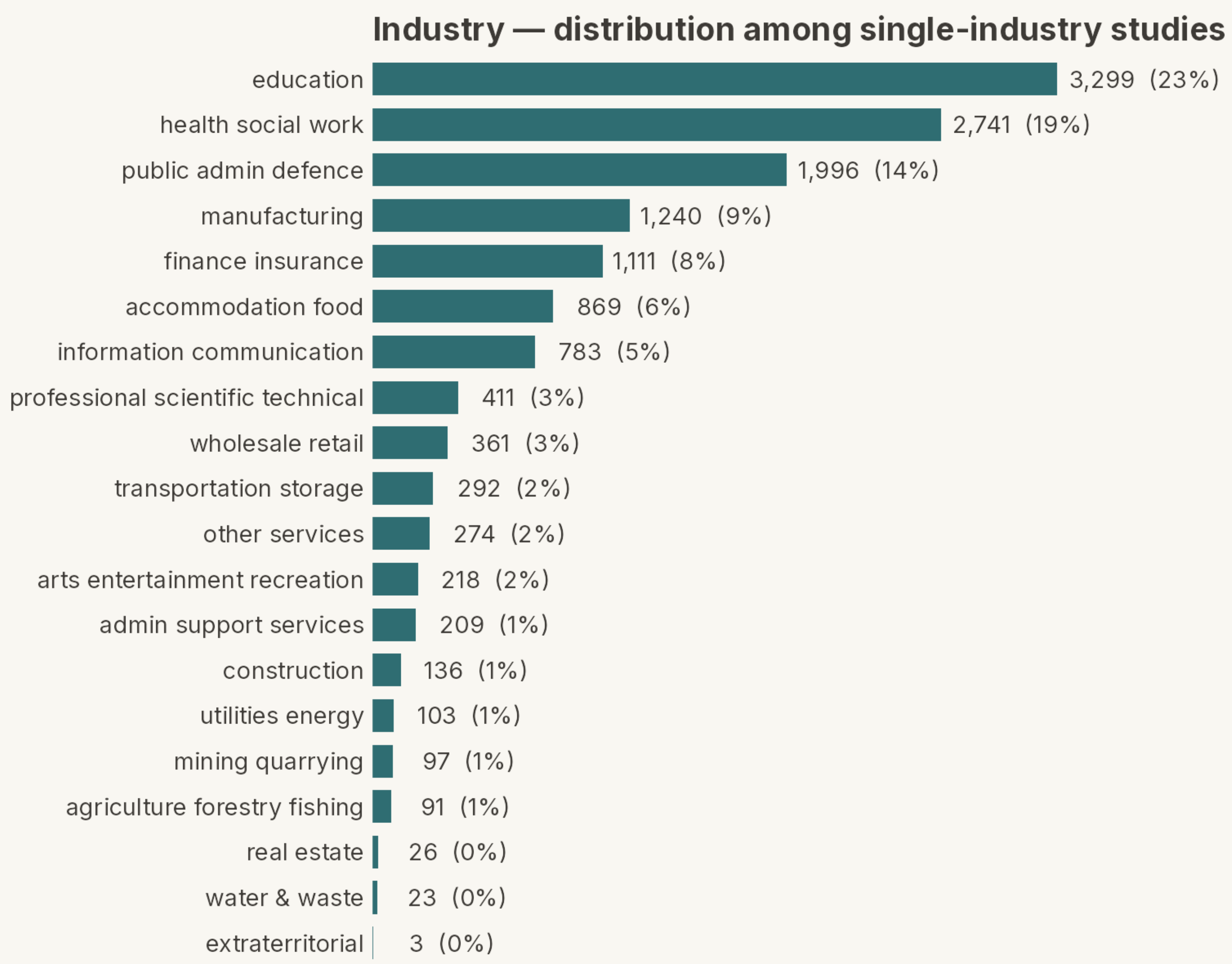
Full coded sample, N = 22,484




### Industry taxonomy — ISIC Rev. 4

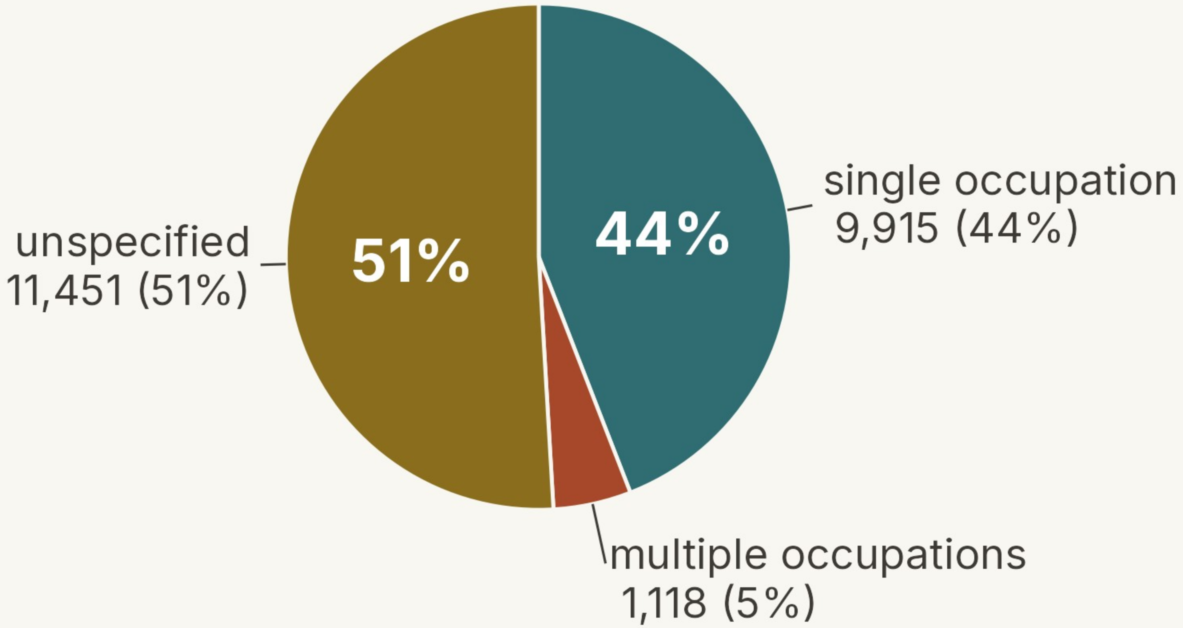
The UN's standard industry taxonomy — 21 letter-coded sections, A (agriculture, forestry & fishing) through U (extraterritorial organizations).

[Taxonomy Details Website]


Industry — distribution among single-industry studies
education
3,299 (23%)
health social work
2,741 (19%)
public admin defence
1,996 (14%)
manufacturing
1,240 (9%)
finance insurance
1,111 (8%)
accommodation food
869 (6%)
information communication
783 (5%)
professional scientific technical
411 (3%)
wholesale retail
361 (3%)
transportation storage
292 (2%)
other services
274 (2%)
arts entertainment recreation
218 (2%)
admin support services
209 (1%)
construction
136 (1%)
utilities energy
103 (1%)
mining quarrying
97 (1%)
agriculture forestry fishing
91 (1%)
real estate
26 (0%)
water & waste
23 (0%)
extraterritorial
3 (0%)

# What Occupations have been Studied

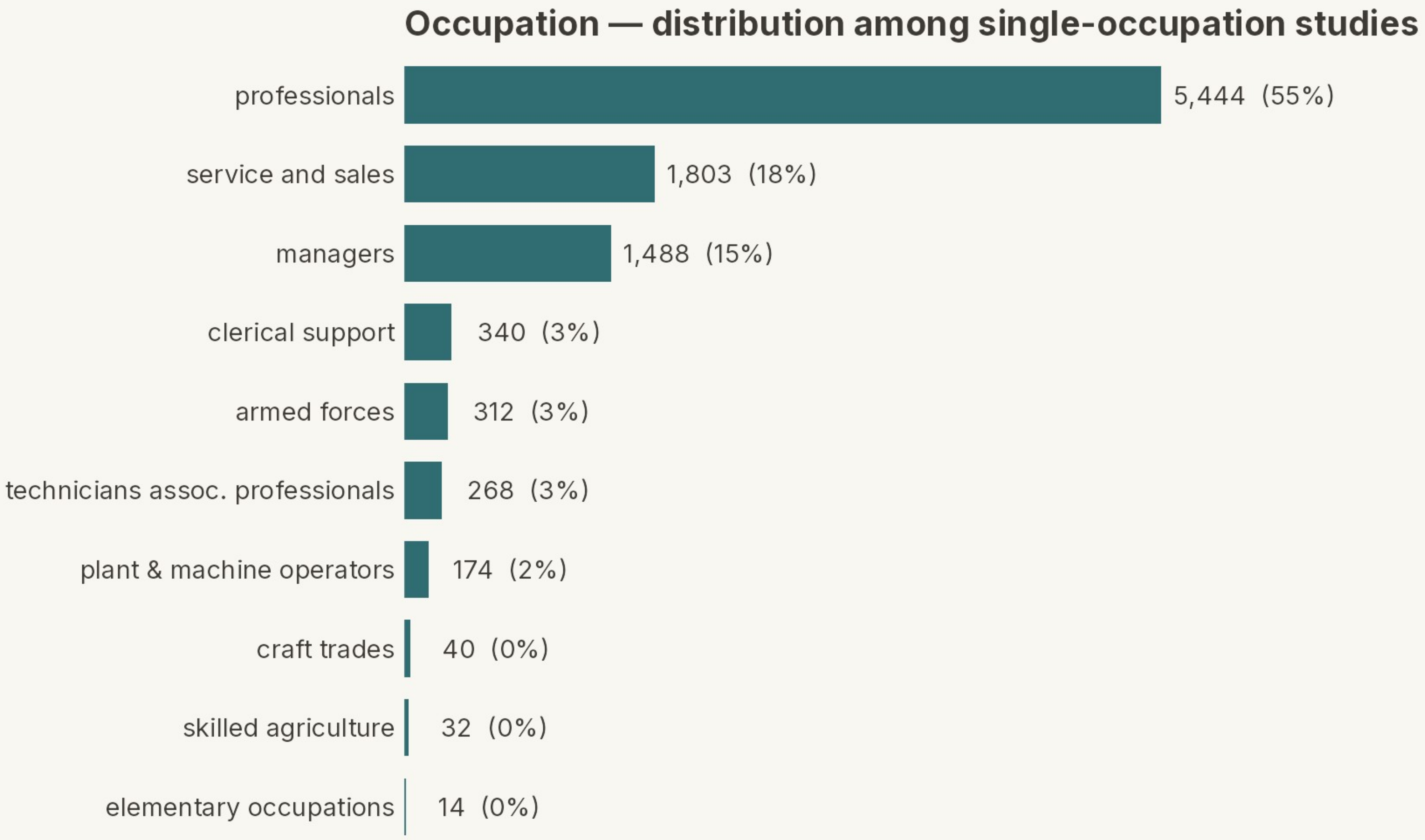
Full coded sample, N = 22,484




### Occupation taxonomy — ISCO-08

The ILO's standard occupation taxonomy (the counterpart to ISIC) — 10 major groups, 0 (armed forces) through 9 (elementary occupations).

[Taxonomy Details Website]


Occupation — distribution among single-occupation studies
professionals
5,444 (55%)
service and sales
1,803 (18%)
managers
1,488 (15%)
clerical support
340 (3%)
armed forces
312 (3%)
technicians assoc. professionals
268 (3%)
plant & machine operators
174 (2%)
craft trades
40 (0%)
skilled agriculture
32 (0%)
elementary occupations
14 (0%)

## How is organizational commitment positioned in the literature?

It plays four primary roles — as a predictor, a criterion, a correlate, and a mediator — and the balance has shifted over time.

**How the four roles were assigned**

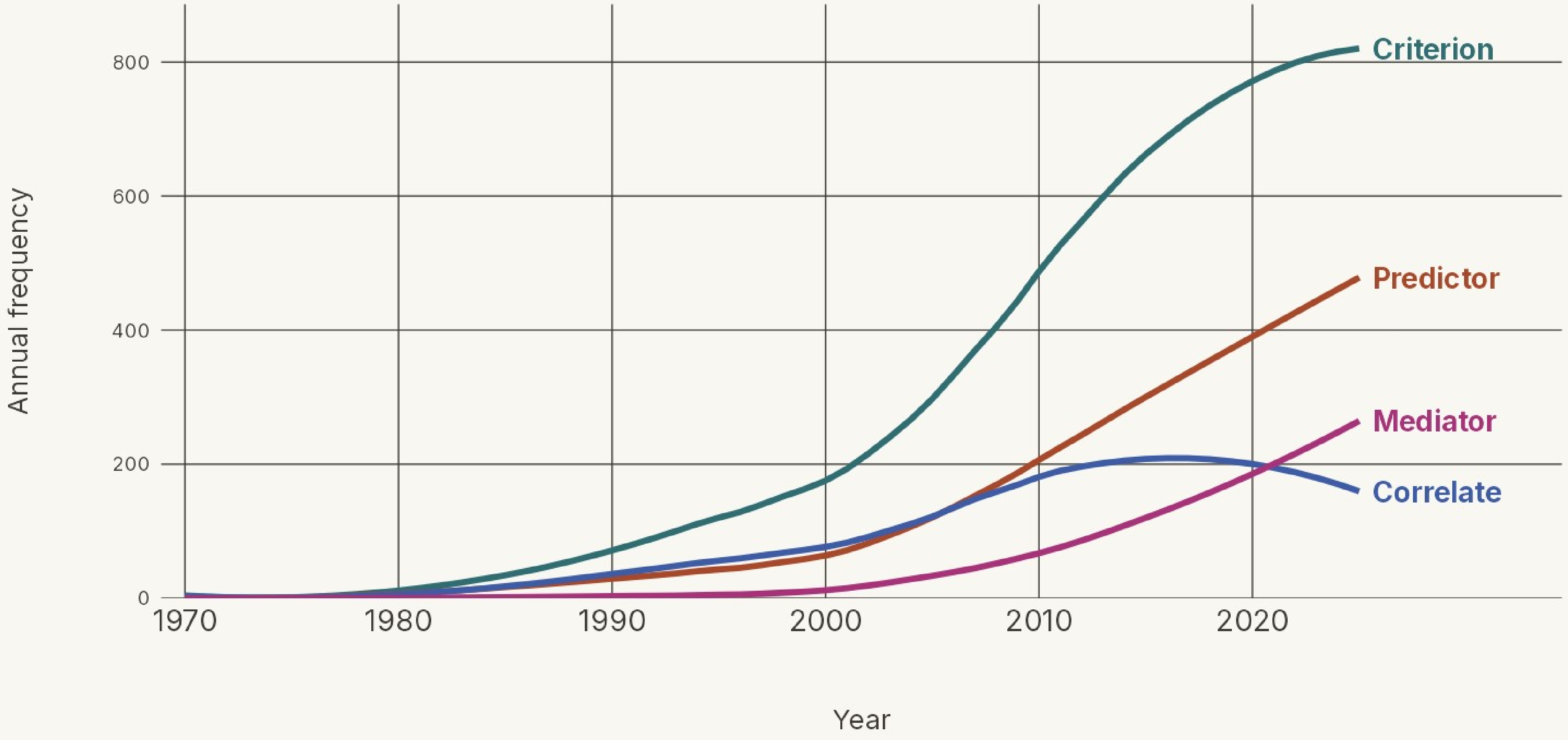
We trained the AI to build a path model from each abstract's text. A variable's position in that path model determined which of the four roles it plays.

## How ‘organizational commitment’ is positioned

Annual count of abstracts using commitment in each role

## How ‘organizational commitment’ is positioned

Annual count of abstracts using commitment in each role

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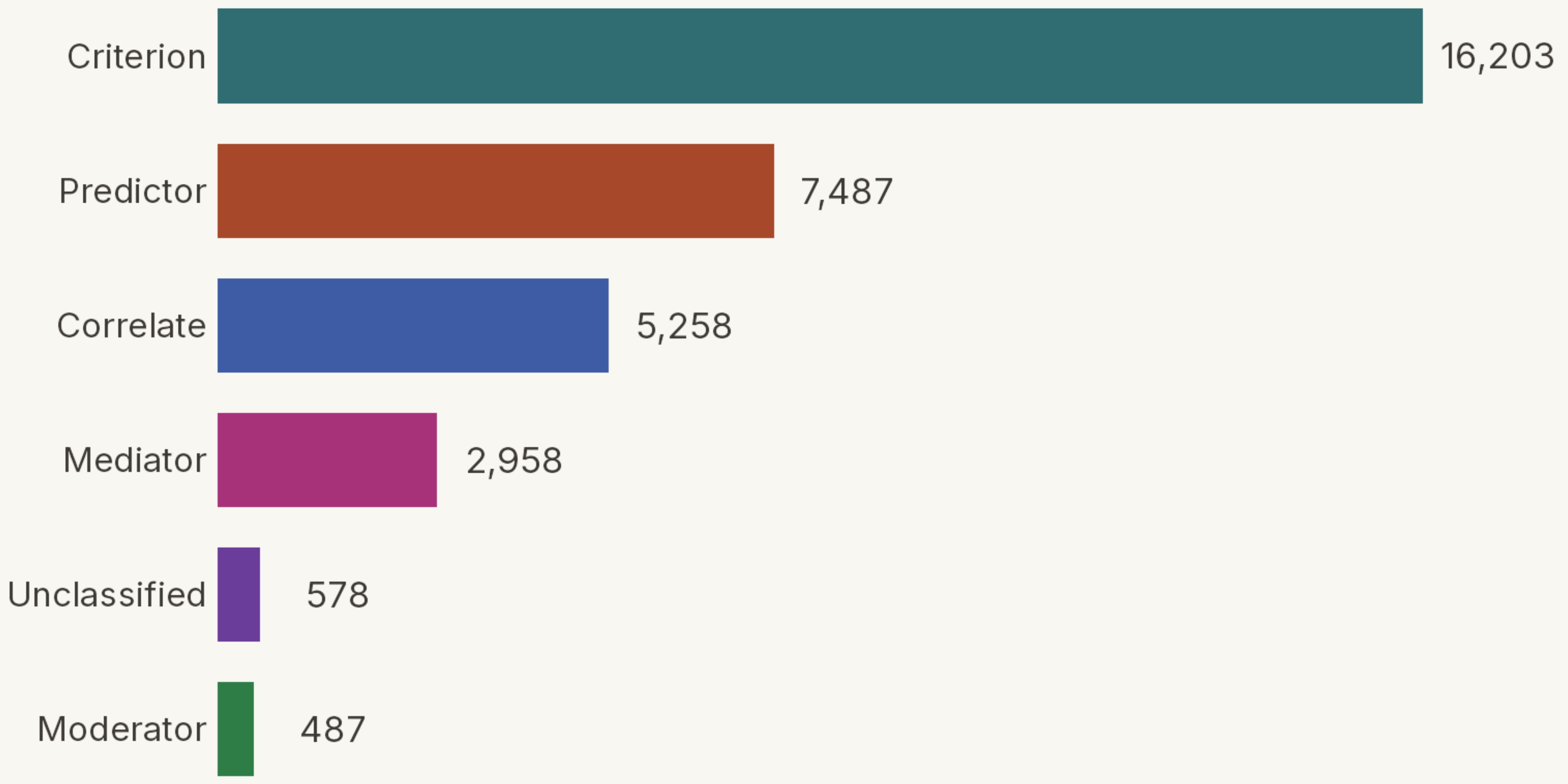
## How ‘organizational commitment’ is positioned — all six roles

Total studies coded in each role — a study can hold several, so bars exceed the 22,484 coded



What variables served as predictors and criteria for commitment?

# How did we identify and combine variable labels

STEP

*1*

### Extract the raw labels

Direct extraction of the coded text, “as is.”

STEP

*2*

### Merge plural / spelling variants

Combine simple plural and spelling variants (e.g., “policies” → “policy”).

STEP

*3*

### Group by meaning

**Predictors:** the AI proposed the families.

**Outcomes:** we supplied them; the AI sorted labels in.

**Steps 2 and 3 by hand would be formidable tasks — AI makes them manageable.**

# How many distinct variables did we extract?

After merging spelling and plural variants — before grouping by meaning

**Step 2**

Merge plural / spelling variants

**Predictors of commitment**

14,390

distinct variables

**Outcomes of commitment**

3,320

distinct variables

# One construct, every language

For every study, the AI extracts the variables it examines — reading the abstract in its own language — and records each under one English label, so studies in any language pool into the same construct.

**As written across languages**

| | |
|---|---|
| job satisfaction | English |
| 工作满意度 | Chinese |
| satisfação no trabalho | Portuguese |
| iş tatmini | Turkish |
| satisfacción laboral | Spanish |
| satisfaction au travail | French |
| 직무 만족 | Korean |
| الرضا الوظيفي | Arabic |

**The unified label**

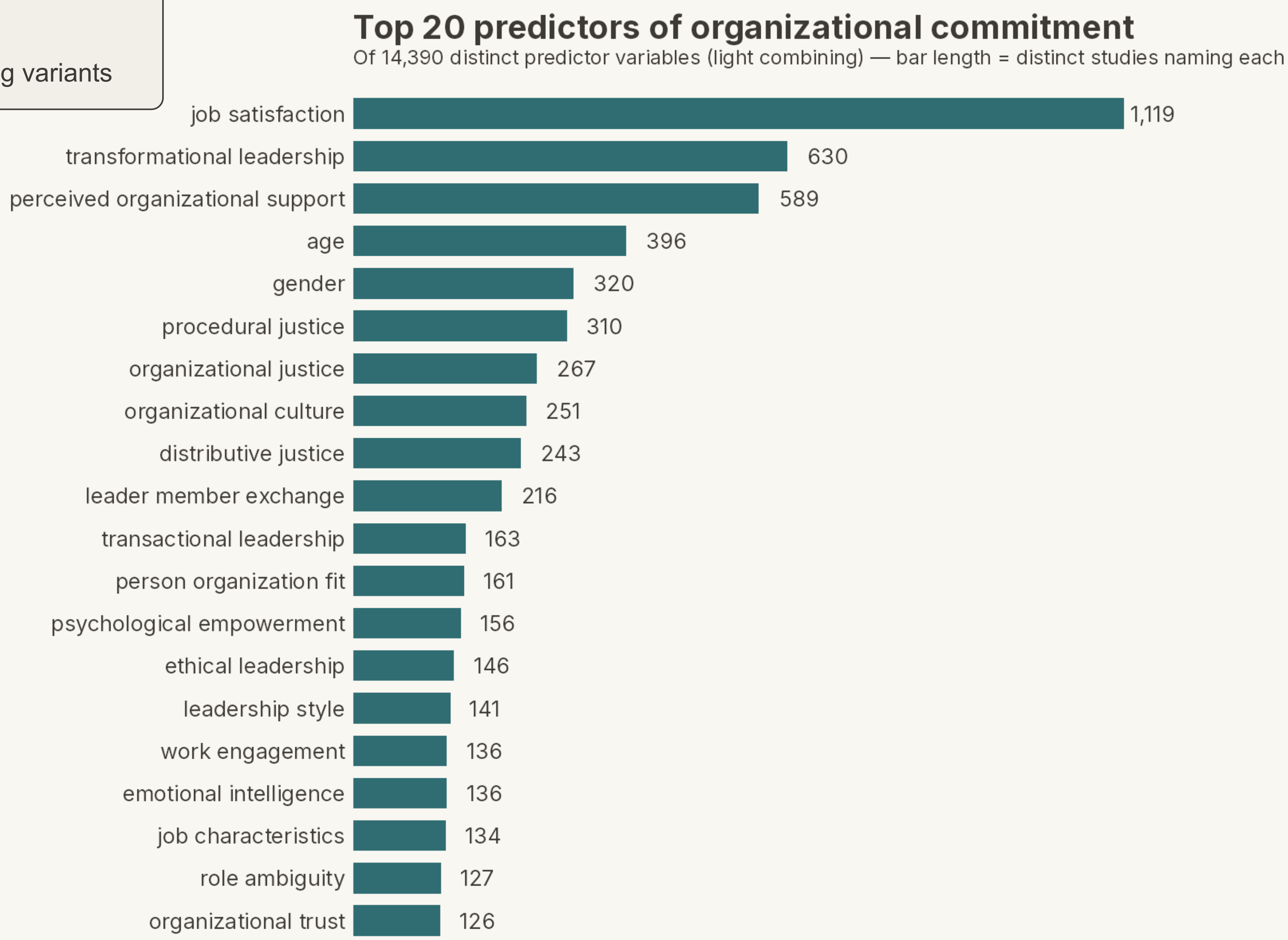
## job_satisfaction

one English label · **2,085** studies pooled

The alternative — human coding — would either **miss the non-English articles** or require **hiring multiple language-proficient coders**. The AI needs neither.

**Step 2**

Merge plural / spelling variants

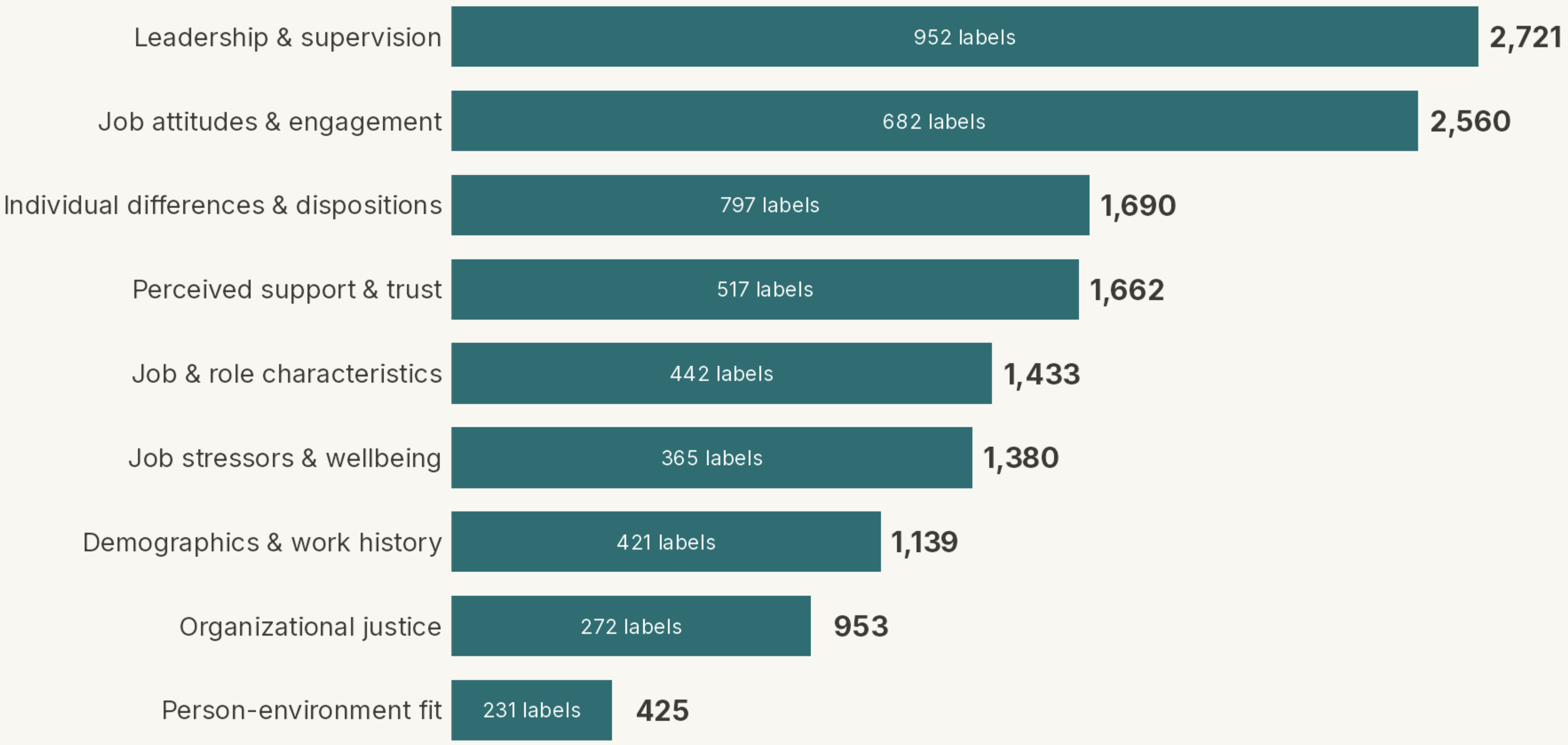


**Step 3**

Group by meaning

## Predictors of organizational commitment, in nine categories

Bar & end label = distinct studies naming ≥1 predictor in the category.
Inside each bar = number of variable labels collapsed into it. (“Other” not shown.)



## Step 2

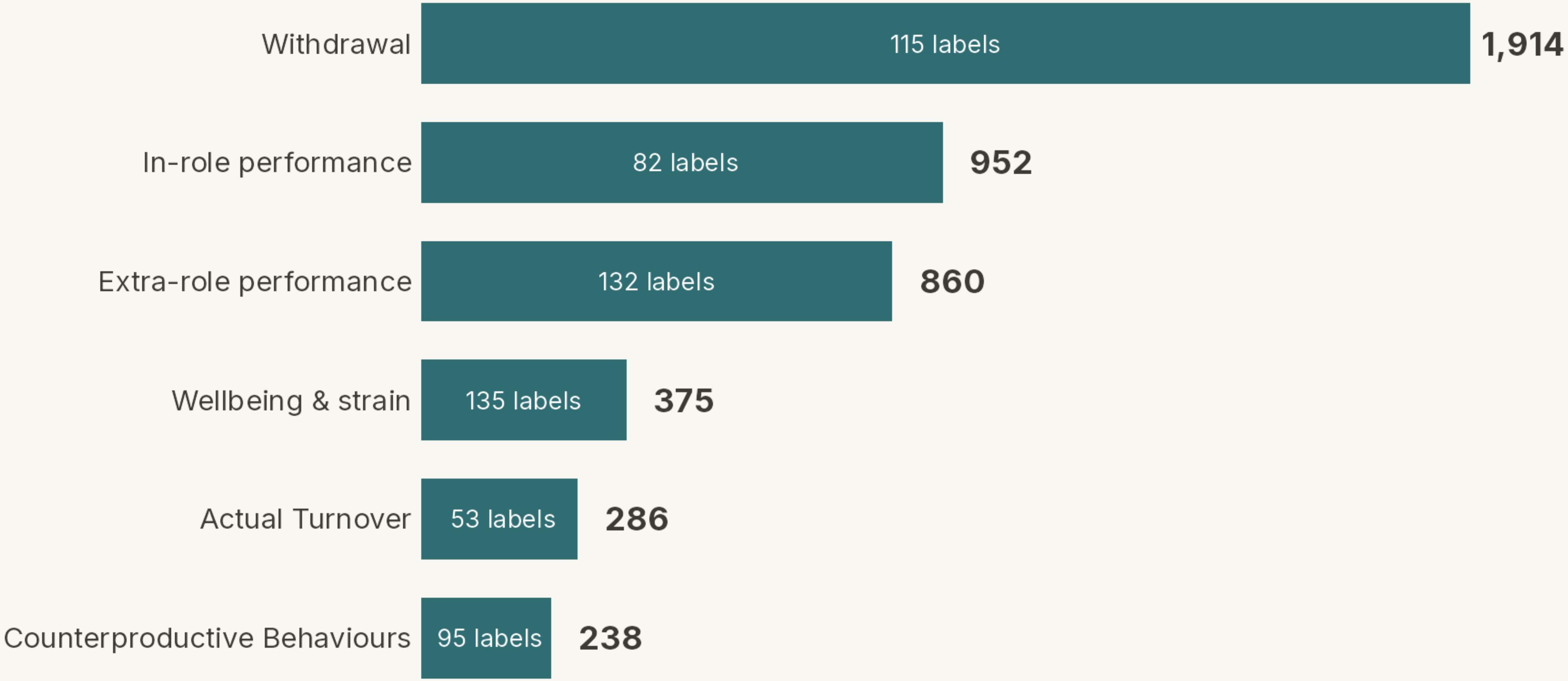
Merge plural / spelling variants

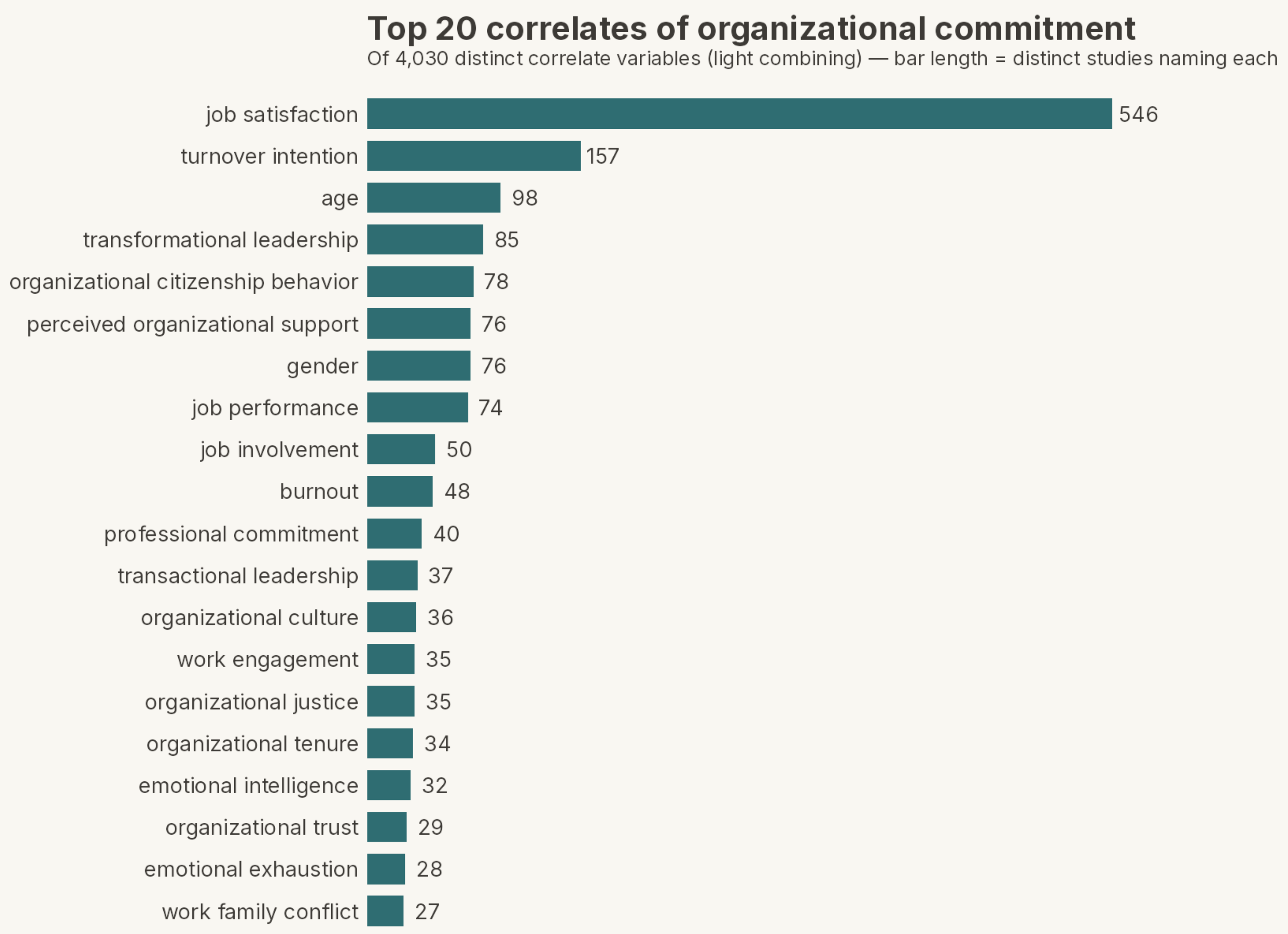
**Step 3**

Group by meaning

## Outcomes of organizational commitment, in six categories

Bar & end label = distinct studies naming ≥1 outcome in the category.
Inside each bar = number of variable labels collapsed into it.




Top 20 correlates of organizational commitment
Of 4,030 distinct correlate variables (light combining) — bar length = distinct studies naming each
job satisfaction 546
turnover intention 157
age 98
transformational leadership 85
organizational citizenship behavior 78
perceived organizational support 76
gender 76
job performance 74
job involvement 50
burnout 48
professional commitment 40
transactional leadership 37
organizational culture 36
work engagement 35
organizational justice 35
organizational tenure 34
emotional intelligence 32
organizational trust 29
emotional exhaustion 28
work family conflict 27

FROM ABSTRACTS TO ANALYSIS

# *Next Steps*

The work ahead, in four steps.

# How accurate is AI coding? — from prior research

## Relative Coding Accuracy

**97.0%**
AI

**92.1%**
expert humans

of extracted values confirmed correct
4 meta-analyses · 632 studies · 20,476 data points

## Time saving with AI

**≈ 32×**

faster than coding by hand

130 hours of hand coding, done by the AI in about 4 hours.
One meta-analysis: 12,025 numbers from 390 studies · one coder.

## How the accuracy was measured

The AI and the human team each extracted every value independently. Matches were counted correct; disagreements were re-checked against the source article by an expert to decide which side was right. Each method's accuracy is the percentage of values confirmed correct, averaged across the four datasets.

Mehr, A., Howard, J. L., Nouroozi, C., Khorramnazari, B., Banks, G. C., Tay, L., Cuijpers, P., Miguel, C., Harrer, M., Meyer, J. P., Stanley, D. J., Wang, X., Luo, G., Huo, B., Liu, J., & Rousseau, D. M. (2026). *Improving Evidence Synthesis with Artificial Intelligence* [Preprint].

# Next steps

From the coded abstracts to a completed meta-analysis — four steps from here.

1 **Download the articles**

Retrieve the full texts behind the coded abstracts.

2 **Code the articles**

Extract effect sizes and study details from each.

3 **Formalize the variable taxonomies**

Finalize the construct groupings from these results.

4 **Begin analyses**

Run the meta-analytic models.

SCAN TO VIEW & SHARE THESE SLIDES

**tinyurl.com/icap2026-commitment**

The download includes a Methodological Deep Dive we didn't have time to present.

# *Toward an AI-assisted Comprehensive Meta-analysis of the Organizational Commitment Literature*

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the SYNTHESIS COMPANY of CALIFORNIA

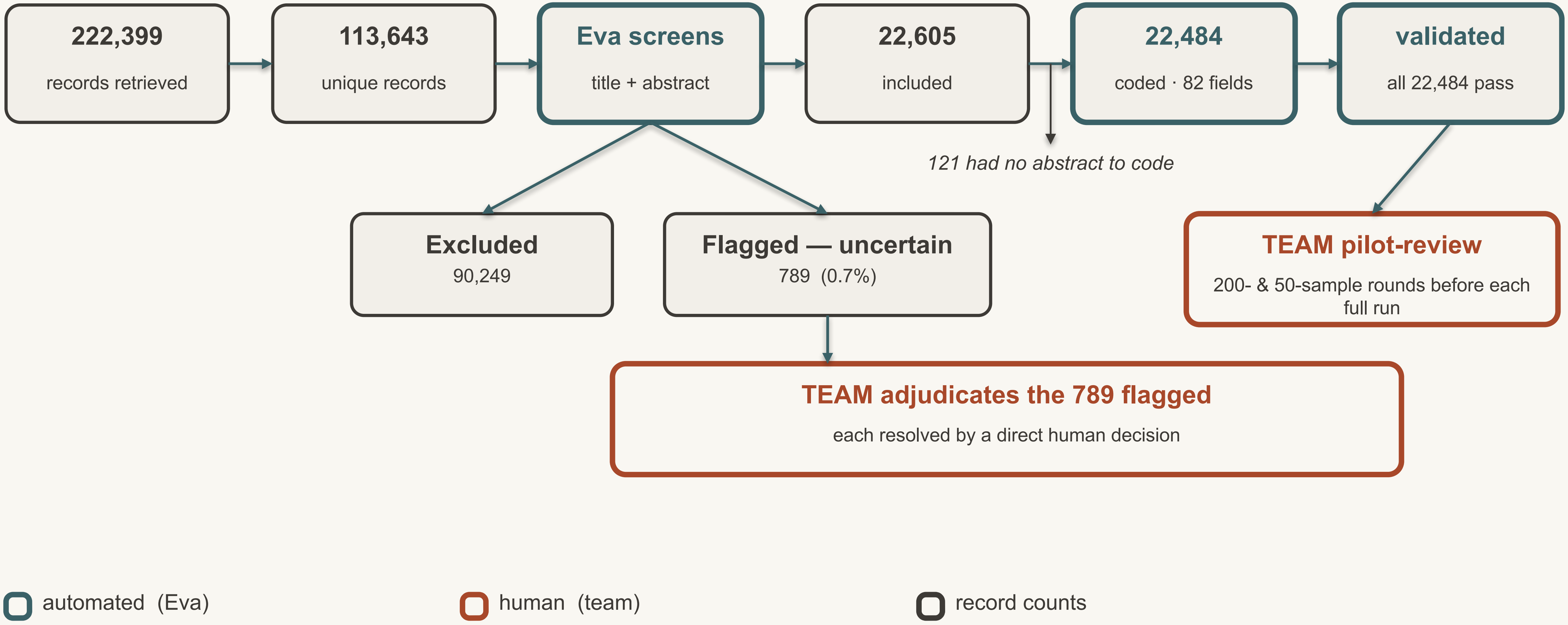
FOR READERS OF THE DOWNLOADED SLIDES

# *Methodological Deep Dive*

The pipeline, calibration, coding scheme, and full counts — detail we skip in the talk.

# The workflow, end to end

Every record's path — the fan-out at screening, and the human decisions, made explicit



# The pipeline in full

Every record's path from retrieval to coded abstract — human steps marked in brown

| | | |
|---|---|---|
| *1* | **Retrieve** | 222,399 records — 7 databases, keyword + cited-reference searches |
| *2* | **De-duplicate** | 113,643 unique records after removing overlapping hits |
| *3* | **Screen** | Eva reads each record's title + abstract and applies 4 inclusion criteria — primary empirical, quantitative, organizational context, commitment measured: 22,605 include · 90,249 exclude · 789 flagged uncertain (0.7%) |
| *4* | **Adjudicate** | Team resolves the 789 cases the AI flagged as uncertain — each gets a direct human decision (the AI never guesses on them) |
| *5* | **Code** | 22,484 abstracts (121 of the 22,605 relevant records had no abstract to code) — 82 structured fields each, with a verbatim quote per field |
| *6* | **Validate** | All 22,484 pass automated consistency checks; team pilot-reviews before each full run |

# Calibration & human oversight

Humans ran a blind spot-check, then concentrated effort on the cases the AI itself flagged as uncertain

| STAGE | WHEN | WHAT HAPPENED |
|---|---|---|
| **Initial alignment** | Sep 2025 | 40 abstracts blind-reviewed — 39/40 agreement (the one mismatch was a case Eva had flagged uncertain). Adopted the rule: when in doubt, err toward inclusion. |
| **Uncertain-case calibration** | Jan–Feb 2026 | 195 records from ~5,400 uncertain cases, blind-reviewed by three team members. Disagreements traced to undefined rules (missing / non-English abstracts) — the reviewers disagreed too. Rules made explicit; the whole set re-screened → uncertain cases fell from ~5,400 to 789. |
| **Uncertainty adjudication** | Feb–Mar 2026 | Team hand-reviewed the 789 cases the AI flagged as uncertain — each resolved by a direct human decision rather than an AI guess. |
| **Coding pilot** | May–Jun 2026 | Codebook built with the team; a 50-sample review + async re-test surfaced 5 fixes (e.g., a mandatory path-diagram step, a level-of-analysis field). Only then the full 22,484 run. |

**Net: roughly 1% of the 113,643 records ever needed manual screening — concentrated exactly where the abstracts were genuinely ambiguous.**

# What was coded

82 structured fields per abstract — each carrying a verbatim quote as its evidence

**PATH-MODEL FIRST**

Eva drew each study's path model, located commitment within it, and only then assigned commitment's role — predictor, outcome, mediator, or moderator.

### The 82 fields capture

- Commitment role (predictor / outcome / mediator / moderator)
- Commitment type (affective / continuance / normative)
- Level of analysis (individual / unit / organization)
- Country of data collection (4-level evidence hierarchy)
- Industry — ISIC Rev. 4
- Occupation — ISCO-08
- Every predictor, outcome, mediator & moderator variable
- Variable names normalized to standard English (any source language)

---

**Evidence, not assertion** — every field carries a verbatim quote from the abstract to support it.

# The numbers — and where they come from

Counts are verified from this project's database; accuracy figures belong to the validation study

THIS PROJECT — DATABASE-VERIFIED

| | |
|---|---|
| 222,399 | records retrieved |
| 113,643 | unique after de-duplication |
| 22,605 | included · 90,249 excluded |
| 789 | flagged uncertain — all human-adjudicated |
| ~1% | of records needed any manual screening |
| 22,484 | abstracts coded · 82 fields each |
| 39/40 | project blind-check agreement |

FROM THE VALIDATION STUDY — MEHR ET AL.

**Screening**

97.2% / 91.6% vs expert human screeners

**Coding**

97.0% / 92.1% vs expert human coders

**Tested on**

5 meta-analyses · 30,150 screening documents

Sensitivity and specificity are properties of the validation study, not of this meta-analysis. This project reports only its own database-verified counts and the 39/40 blind-check.

SEED STUDIES FOR THE CITED-REFERENCE SEARCH

# *Foundational Works*

What we mean by “foundational” — the studies whose citations we followed in the cited-reference search.

# Foundational works



The cited-reference search retrieved every record citing any of these 14 works

**Allen, N. J., & Meyer, J. P. (1990).** The measurement and antecedents of affective, continuance and normative commitment to the organization. *Journal of Occupational Psychology, 63*(1), 1–18.

**Allen, N. J., & Meyer, J. P. (1996).** Affective, continuance, and normative commitment to the organization: An examination of construct validity. *Journal of Vocational Behavior, 49*(3), 252–276.

**Cook, J., & Wall, T. (1980).** New work attitude measures of trust, organizational commitment and personal need non-fulfilment. *Journal of Occupational Psychology, 53*(1), 39–52.

**Klein, H. J., Cooper, J. T., Molloy, J. C., & Swanson, J. A. (2014).** The assessment of commitment: Advantages of a unidimensional, target-free approach. *Journal of Applied Psychology, 99*(2), 222–238.

**Meyer, J. P., & Allen, N. J. (1984).** Testing the “side-bet theory” of organizational commitment: Some methodological considerations. *Journal of Applied Psychology, 69*(3), 372–378.

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